

- Program - The current population of a town is 10000. The population of the town is increasing at the rate of 10% per year. You have to write a program to find out the population at the end of each of the last 10 years.

```
# Code here
curr_pop = 10000

for i in range(10,0,-1):
    print(i,curr_pop)
    curr_pop = curr_pop/1.1
```

```
10 10000
9 9090.90909090909
8 8264.462809917353
7 7513.148009015775
6 6830.134553650703
5 6209.213230591548
4 5644.739300537771
3 5131.5811823070635
2 4665.07380209733
1 4240.976183724845
```

- Sequence sum

$1/1! + 2/2! + 3/3! + \dots$

```
# Code here

n = int(input('enter n'))

result = 0
fact = 1

for i in range(1,n+1):
    fact = fact * i
    result = result + i/fact

print(result)
```

```
enter n2
2.0
```

- Nested Loops

```
# Examples -> unique pairs

for i in range(1,5):
    for j in range(1,5):
        print(i,j)
```

```
1 1
1 2
1 3
1 4
2 1
2 2
2 3
2 4
3 1
3 2
3 3
3 4
4 1
4 2
4 3
4 4
```

▼ Pattern 1

```
***
****
***
```

```
# code here

rows = int(input('enter number of rows'))

for i in range(1,rows+1):
    for j in range(1,i+1):
        print('*',end='')
    print()
```

```
enter number of rows10
*
**
***
****
*****
*****
*****
*****
*****
*****
```

▼ Pattern 2

```
1
121
12321
1234321
```

```
# Code here
rows = int(input('enter number of rows'))

for i in range(1,rows+1):
    for j in range(1,i+1):
        print(j,end='')
    for k in range(i-1,0,-1):
        print(k,end='')

    print()
```

```
enter number of rows4
1
121
12321
1234321
```

▼ Loop Control Statement

- Break
- Continue
- Pass

```
for i in range(1,10):
    if i == 5:
        break
    print(i)
```

```
1
2
3
4
```

```
lower = int(input('enter lower range'))
upper = int(input('enter upper range'))

for i in range(lower,upper+1):
    for j in range(2,i):
        if i%j == 0:
            break
    else:
        print(i)
```

```
enter lower range10
enter upper range100
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
```

```
# Continue
for i in range(1,10):
    if i == 5:
        continue
    print(i)
```

```
1
2
3
4
6
7
8
9
```

```
for i in range(1,10):
    pass
```

Strings are sequence of Characters

In Python specifically, strings are a sequence of Unicode Characters

- Creating Strings
- Accessing Strings
- Adding Chars to Strings
- Editing Strings
- Deleting Strings
- Operations on Strings
- String Functions

✓ Creating Stings

```
s = 'hello'
s = "hello"
# multiline strings
s = '''hello'''
s = """hello"""
s = str('hello')
print(s)
```

```
hello
```

```
"it's raining outside"
```

```
'it's raining outside'
```

✓ Accessing Substrings from a String

```
# Positive Indexing
s = 'hello world'
print(s[41])
```

```
-----
IndexError                                Traceback (most recent call last)
<ipython-input-61-633ba99ed6e5> in <module>
      1 # Positive Indexing
      2 s = 'hello world'
----> 3 print(s[41])

IndexError: string index out of range
```

```
# Negative Indexing
s = 'hello world'
print(s[-3])
```

```
r
```

```
# Slicing
s = 'hello world'
print(s[6:0:-2])
```

```
wol
```

```
print(s[::-1])
```

```
dlrow olleh
```

```
s = 'hello world'
print(s[-1:-6:-1])
```

```
dlrow
```

✓ Editing and Deleting in Strings

```
s = 'hello world'
s[0] = 'H'

# Python strings are immutable
```

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-80-0c8a824e3b73> in <module>
      1 s = 'hello world'
----> 2 s[0] = 'H'

TypeError: 'str' object does not support item assignment
```

```
s = 'hello world'
del s
print(s)
```

```
-----
NameError                                Traceback (most recent call last)
<ipython-input-81-9ae37fbf1c6c> in <module>
      1 s = 'hello world'
      2 del s
----> 3 print(s)

NameError: name 's' is not defined
```

```
s = 'hello world'
del s[-1:-5:2]
print(s)
```

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-82-d0d823eafb6b> in <module>
      1 s = 'hello world'
----> 2 del s[-1:-5:2]
      3 print(s)

TypeError: 'str' object does not support item deletion
```

Operations on Strings

- Arithmetic Operations
- Relational Operations
- Logical Operations
- Loops on Strings
- Membership Operations

```
print('delhi' + ' ' + 'mumbai')
```

```
delhi mumbai
```

```
print('delhi'*5)
```

```
delhidelhidelhidelhidelhi
```

```
print("""*50
```

```
*****
```

```
'delhi' != 'delhi'
```

```
False
```

```
'mumbai' > 'pune'
# lexicographically
```

```
False
```

```
'Pune' > 'pune'
```

```
False
```

```
'hello' and 'world'
```

```
'world'
```

```
'hello' or 'world'
```

```
'hello'
```

```
'' and 'world'
```

```
''
```

```
'' or 'world'
```

```
'world'
```

```
'hello' or 'world'
```

```
'hello'
```

```
'hello' and 'world'
```

```
'world'
```

```
not 'hello'
```

```
False
```

```
for i in 'hello':  
    print(i)
```

```
h  
e  
l  
l  
o
```

```
for i in 'delhi':  
    print('pune')
```

```
pune  
pune  
pune  
pune  
pune
```

```
'D' in 'delhi'
```

```
False
```

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▼ Common Functions

- len
- max
- min
- sorted

```
len('hello world')
```

```
11
```

```
max('hello world')
```

```
'w'
```

```
min('hello world')
```

```
' '
```

```
sorted('hello world',reverse=True)
```

```
['w', 'r', 'o', 'o', 'l', 'l', 'l', 'h', 'e', 'd', ' ']
```

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✓ Capitalize/Title/Upper/Lower/Swapcase

```
s = 'hello world'
print(s.capitalize())
print(s)
```

```
Hello world
hello world
```

```
s.title()
```

```
'Hello World'
```

```
s.upper()
```

```
'HELLO WORLD'
```

```
'Hello Wolrd'.lower()
```

```
'hello wolrd'
```

```
'HeLlO WorLD'.swapcase()
```

```
'hELLo wORld'
```

✓ Count/Find/Index

```
'my name is nitish'.count('i')
```

```
3
```

```
'my name is nitish'.find('x')
```

```
-1
```

```
'my name is nitish'.index('x')
```

```
-----
ValueError                                Traceback (most recent call last)
<ipython-input-121-12e2ad5b75e9> in <module>
----> 1 'my name is nitish'.index('x')

ValueError: substring not found
```

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✓ endswith/startswith

```
'my name is nitish'.endswith('sho')
```

```
False
```

```
'my name is nitish'.startswith('lmy')
```

```
False
```

✓ format

```
name = 'nitish'  
gender = 'male'
```

```
'Hi my name is {1} and I am a {0}'.format(gender,name)
```

```
'Hi my name is nitish and I am a male'
```

✓ isalnum/ isalpha/ isdigit/ isidentifier

```
'nitish1234%'.isalnum()
```

```
False
```

```
'nitish'.isalpha()
```

```
True
```

```
'123abc'.isdigit()
```

```
False
```

```
'first-name'.isidentifier()
```

```
False
```

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✓ Split/Join

```
'hi my name is nitish'.split()
```

```
['hi', 'my', 'name', 'is', 'nitish']
```

```
" ".join(['hi', 'my', 'name', 'is', 'nitish'])
```

```
'hi my name is nitish'
```

✓ Replace

```
'hi my name is nitish'.replace('nitisrgewrhgh','campusx')
```

```
'hi my name is nitish'
```

✓ Strip

```
'nitish'.strip()
```

```
'nitish'
```

✓ Example Programs

```
# Find the length of a given string without using the len() function  
  
s = input('enter the string')  
  
counter = 0
```



```
for i in s:
    counter += 1

print('length of string is',counter)
```

```
enter the stringnitish
length of string is 6
```

```
# Extract username from a given email.
# Eg if the email is nitish24singh@gmail.com
# then the username should be nitish24singh
```

```
s = input('enter the email')

pos = s.index('@')
print(s[0:pos])
```

```
enter the emailsupport@campusx.in
support
```

```
# Count the frequency of a particular character in a provided string.
# Eg 'hello how are you' is the string, the frequency of h in this string is 2.
```

```
s = input('enter the email')
term = input('what would like to search for')

counter = 0
for i in s:
    if i == term:
        counter += 1

print('frequency',counter)
```

```
enter the emailhi how are you
what would like to search foro
frequency 2
```

```
# Write a program which can remove a particular character from a string.
s = input('enter the string')
term = input('what would like to remove')
```

```
result = ''

for i in s:
    if i != term:
        result = result + i

print(result)
```

```
enter the stringnitish
what would like to removei
ntsh
```

```
# Write a program that can check whether a given string is palindrome or not.
# abba
# malayalam
```

```
s = input('enter the string')
flag = True
for i in range(0,len(s)//2):
    if s[i] != s[len(s) - i -1]:
        flag = False
        print('Not a Palindrome')
        break

if flag:
    print('Palindrome')
```

```
enter the stringpython
Not a Palindrome
```

```
# Write a program to count the number of words in a string without split()

s = input('enter the string')
L = []
temp = ''
for i in s:

    if i != ' ':
        temp = temp + i
    else:
        L.append(temp)
        temp = ''

L.append(temp)
print(L)
```

```
enter the stringhi how are you
['hi', 'how', 'are', 'you']
```

```
# Write a python program to convert a string to title case without using the title()
s = input('enter the string')

L = []
for i in s.split():
    L.append(i[0].upper() + i[1:].lower())

print(" ".join(L))
```

```
enter the stringhi my name is NitiSh
Hi My Name Is Nitish
```

```
# Write a program that can convert an integer to string.

number = int(input('enter the number'))

digits = '0123456789'
result = ''
while number != 0:
    result = digits[number % 10] + result
    number = number//10

print(result)
print(type(result))
```

```
enter the number345
345
<class 'str'>
```

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